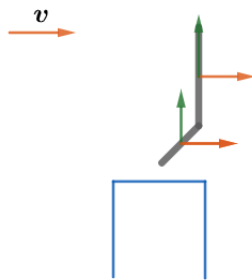


2021 F=ma Exam: Problem 18

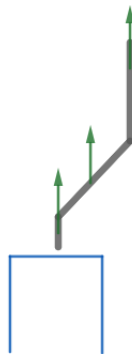
Kevin S. Huang



Initially the smoke rises upward since the air around it is still.



When the wind blows to the right, the smoke already in the air gains a rightward component of velocity so that relative to the wind, it is still rising with constant speed. New smoke also moves in this direction.



When the wind stops, all the smoke rises upward just like in the beginning. Thus, the answer is B.